PRODUCT BLUEPRINT | PROPTECH

Gated Community Platform

One modular, white-label platform for the full property lifecycle: pre-sale to daily community living, residential and commercial, with an AI layer on top.

Build overview, architecture, modules, AI layer, version management and roadmap

[Your company name] | October 2026 | Version 1.0 draft

1. Vision and scope

One modular platform for the full property lifecycle, from pre-sale to daily community living, with an AI layer on top. Developers, owners, tenants, guards, agents and admins each get the right tools. Every client gets its own brand and domain, every capability is a plug-in module, and client-specific needs are added as isolated plugins that never touch existing modules.

- **Who it serves:** large developers and their communities, residential and commercial (offices, retail, mixed-use).
- **Lifecycle covered:** pre-sale, post-sale and handover, tenancy, daily operations, access and billing.
- **Approach:** every capability is a separate module, so a client switches on only what it needs.
- **White label:** each client runs under its own brand and domain, configured without code changes.
- **Safe customisation:** special requirements become extension plugins, so other clients are never affected.
- **AI where it helps:** natural-language help, routine-work automation, proactive alerts, and an agent that recommends the right modules.

Result in one line: a new client goes live on its own brand and domain through configuration, a special request becomes an isolated plugin, and upgrades happen module by module with a rollback path.

2. Industry challenges we solve

Area	Challenge today	Our answer
Presales	Leads scattered across brokers and portals; commission disputes; slow booking workflow	CRM pipeline, agent portal, booking and commission tracking
Post-sales	Handover delays, paper snag lists, chasing documents and demand letters	Milestone billing, photo snagging, document vault, handover checklist
Security	Paper registers, unverified visitors, no guard audit trail	Pre-approved passes, guard app, logs, SOS and incident reports
Access	RFID, intercom and CCTV that do not talk to each other	Smart access plugin with hardware adapters
Billing	Manual maintenance collection, late dues, opaque utility charges	Auto invoicing, online payments, reminders, ledger
Commercial	Fit-out, CAM charges, visitor flow and SLAs run in spreadsheets	Lease, CAM, access and helpdesk with SLA timers

3. Market scan: what exists today

Existing products each cover part of the journey. Products named below are examples found in public listings; verify current features before quoting them to a client.

Region	Examples	Typical features
UAE	Al-Futtaim Living, MAG City app, Emaar One, Danube, DHRE RealConnect, NoBrokerHood UAE	Maintenance requests, guest approvals, amenity booking, move-in/out, NOC, payments, Mollak integration, manager portal
India	MyGate, NoBrokerHood, ADDA, ApnaComplex	Visitor and parcel approval, staff management, society billing with GST/TDS, helpdesk, polls, marketplace, tanker monitoring
Singapore, Malaysia, China	Habitap, Dahua, Hikvision solutions	Face, QR and plate-recognition access, video intercom, concierge, facility booking, building-system integration
Europe	Verwaltio, ImmoDesk, VecinApp	Rent tracking, tenant portal, meter readings, rent indexation, dunning, SEPA, voting, GDPR hosting

Our gap: few products connect presale, handover, tenants, access and billing for both residential and commercial, with white labeling, plug-in customisation and AI agents, and with local compliance built in (Mollak, GST, SEPA, GDPR, UAE PDPL, India DPDP).

4. Users and what each one gets

User	Main needs	Key features
Prospect / buyer	Explore, decide, book	Project info, unit availability, payment plans, site-visit booking, lead capture
Owner	Track property and dues	Possession and snag tracking, dues and receipts, documents, bylaw answers
Tenant	Live smoothly	Onboarding and move-in checklist, agreement and rent queries, amenity booking, complaints
Guard / receptionist	Fast gate decisions	Pass check, entry log, deliveries, incident reports, local-language voice
Committee / admin	Run the community	Complaint routing, notices, finance and meeting summaries, anomaly alerts
Builder / sales team	Sell and support	Inventory, quotes, follow-ups, lead summaries, dashboards
Agent / broker	Close deals	Inventory view, bookings, commission tracking, licence verification
Vendor / staff	Do the work	Work orders, completion proof with photos
Commercial occupier	Operate an office or shop	Staff access cards, visitor pre-registration, lease and CAM invoices, tickets
Platform admin	Operate the platform	Client onboarding, branding, modules, versions, audit logs

Also supported: co-owner and family members, domestic staff, delivery partners, auditors (read-only) and security supervisors.

5. Platform modules

Each module is an independent plugin with its own routes, tables, permissions and tests. Modules talk through events and shared contracts.

Module	What it does	Phase
Pre-sale and agents	Project catalogue, inventory, pricing and payment plans, CRM pipeline, site visits, broker portal, commissions	4
Post-sale	Booking to possession journey, demand letters, documents, snagging, handover, ownership records, resale and NOC	4
Tenant management	Onboarding with KYC, agreements, owner NOC, rent collection, move-in and move-out	1
Visitor and guard	Pre-approvals, QR or OTP entry, deliveries, domestic staff passes, guard app, patrols, SOS, logs	1
Smart access	Gate, boom barrier, intercom, face and plate recognition, lift linkage, time-bound access rules	3
Smart billing	Maintenance and utility billing, CAM, receipts, online payments, reminders, ledger, regulatory integrations	3
Helpdesk and facility	Tickets, SLA tracking, vendor work orders, preventive maintenance, asset register	1
Amenities and community	Amenity booking, parking, notices, polls and voting, events, marketplace, documents	3
Commercial	Leases, fit-out approvals, CAM charges, staff access, meeting rooms, energy and compliance reports	5
Admin and analytics	Roles and permissions, reports, dashboards, audit logs, open API and webhooks	1 and 5
AI layer	Role-aware assistants, agents, module advisor (section 8)	2 onward

6. Platform foundations: white label, plug-and-play, versions

These capabilities make the platform plug and play. A new client, a new domain or a new requirement should be a configuration or a plugin, never a rewrite.

6.1 White labeling and custom domains

- **Brand per client:** logo, colours, fonts, app name, email/SMS/WhatsApp templates and default language, all set from an admin panel.
- **Any domain:** a client uses its own domain or sub-domain (for example resident.clientbrand.com). The platform maps domain to client, issues SSL automatically, and falls back to a default sub-domain.
- **Feature switches:** per client and per project, choose which modules are on, which roles exist and which workflows apply.
- **Branded mobile apps:** app name, icon and theme come from the same client configuration.
- **Data isolation:** one codebase with each client's data separated (shared database with strict scoping, or a dedicated database for large clients).
- **Regional settings:** language and right-to-left layouts, currency, tax rules and payment providers set per client or country.

6.2 Plug-and-play modules and safe customisation

- **Module manifest:** every module declares name, version, dependencies, permissions, routes, events, settings and supported core versions.
- **Install, enable, disable per client:** adding or removing a module needs no change to other modules.
- **Loose coupling:** modules communicate through events and defined contracts, never by reading each other's tables or code.
- **Custom requirements as extension plugins:** a client-specific need is its own plugin that hooks into events and settings. Existing modules are not edited, so other clients are never affected.
- **Own data and tests:** each module has prefixed tables, its own migrations and an automated test suite.

6.3 Module catalogue and AI module advisor

All modules and plugins sit in one searchable catalogue so teams find what they need without asking developers.

- **Search and filter:** keyword, category, user role, residential or commercial, integrations, version and status.
- **Each listing shows:** purpose, screenshots, dependencies, settings, changelog, compatibility and which clients use it.
- **Module advisor agent:** describe a need in plain words, for example "visitor approval with QR codes for a commercial office tower". The agent searches the catalogue (RAG over manifests and docs), recommends best-fit modules, checks dependencies and compatibility, and flags gaps that need a custom plugin.
- **With approval it can act:** enable modules on a test environment or raise a custom-module request, using MCP tools such as search_modules, get_module_details, check_compatibility, enable_module and request_custom_module.

6.4 Version management

Area	How we handle it
Versioning	Semantic versioning (MAJOR.MINOR.PATCH) for the core and every module
Compatibility	Manifest lists supported core range and dependency ranges; a checker blocks incompatible installs and upgrades
Per-client pinning	Each client stays on chosen versions and upgrades at its own pace; release channels: stable, beta, long-term support
Safe upgrades	Staging first, automated tests, reversible migrations, feature flags, gradual rollout, one-click rollback
Stable APIs	Versioned APIs (v1, v2) and a deprecation policy so custom plugins keep working
Traceability	Changelog per release and an audit log of who upgraded what, where and when

7. Laravel plugin architecture

Three layers: a small stable core, domain plugins, and swappable provider adapters.

Layer	Contents	Role
Core kernel (always loaded)	Plugin manager, tenancy and domain mapping, RBAC, event bus, contracts, branding and settings, audit log	Boots plugins, enforces rules, resolves each client by domain
Domain plugins (enabled per client or project)	Presale, Post-sales, Tenants, Visitors, Smart access, Billing, Helpdesk, Amenities, Commercial, plus client extension plugins	Business features; talk only through events and contracts
Provider adapters (swappable per country)	Payments, messaging, KYC and e-sign, access hardware, CRM and ERP	Implement core contracts; changing a provider is one new class

Plugin layout and manifest

```
plugins/billing/
  plugin.json                 name, version, requires, core range
  src/BillingServiceProvider.php
  src/Models  Http  Listeners  Jobs
  database/migrations         tables prefixed billing_
  routes/api.php
  tests/
```

```
{ "name": "billing", "version": "1.2.0",
  "core": ">=1.0 <2.0", "requires": { "tenants": "^1.0" },
  "provider": "Plugins\\Billing\\BillingServiceProvider",
  "permissions": ["billing.view", "billing.collect"],
  "events": { "emits": ["InvoiceIssued"], "listens": ["UnitBooked"] } }
```

Rules

- A plugin may depend only on the core or on another plugin's published events and interfaces.
- Each plugin owns its tables with a prefix; cross-plugin links use IDs, not foreign-key joins.
- A plugin registers its routes, menus, dashboard widgets, permissions and settings through hooks in its service provider.
- Routes load only when the plugin is enabled for that client (middleware check). Enabled state lives in a plugin installation table per client and project.
- CI fails any plugin that imports another plugin's internals. Each plugin has its own tests, and migrations are reversible.
- Use queues for notifications, access logs and payment webhooks; expose a versioned API (/api/v1) and webhooks for builders' ERP and CRM.

Technology choices

Layer	Choice	Notes
Core platform	Laravel, plugin-based	Module = package with own routes, migrations, permissions
Multi-tenancy and domains	Tenant-aware Laravel (e.g. stancl/tenancy)	Branding, config and feature switches resolved per domain
Module packaging	Composer packages from a private registry	Versioned and installable per client; registry stores manifests
Database	MySQL	Source of truth for units, residents, tickets, billing
Web and apps	React front end; mobile apps for guards and residents	Guard app simple and voice friendly
AI services	Python agent service, MCP server, vector DB (pgvector, Qdrant or Chroma)	MCP tools map to existing APIs; nothing bypasses permissions
Channels	WhatsApp, SMS, push, in-app chat, voice	Same agents behind every channel

8. The AI layer: agents, RAG and MCP

Simple picture: AI is the brain, RAG is the library, MCP is the hands, and the agent is the employee who thinks and then does the work.

Part	Job	In our platform
RAG	Finds answers in our own documents so the AI does not guess	Bylaws, agreements, brochures, FAQs, circulars, minutes and resolved tickets in a vector database, filtered by role, project and unit
MCP	A standard way for AI to use our systems	Our APIs exposed as tools: create_ticket, check_visitor, get_dues, book_amenity, notify_technician
Agent	Plans steps, calls tools, checks results, replies	Role-specific agents coordinated by a router agent

Worked example

A tenant writes on WhatsApp at 10 pm: "The lift is not working. Can I book the clubhouse?" The agent sees a complaint and a question and confirms the user is a tenant. RAG finds the bylaw that tenants may book the clubhouse with the owner's NOC. MCP calls create_ticket (lift, Tower B, high priority) and notify_technician. The reply gives the ticket number, says the technician is informed, explains the NOC requirement and offers the next free slot.

Agent	Uses RAG for	Uses MCP tools for
Sales concierge	Brochures, pricing, approvals	Lead capture, site-visit booking, quotes
Owner agent	Bylaws, agreements, FAQs	Dues, receipts, document requests, snag status
Tenant agent	Rules, move-in guide	Tickets, amenity booking, rent queries
Guard agent	Gate procedures	Visitor checks, entry log, incident report
Admin agent	Minutes, circulars, past tickets	Ticket routing, notices, summaries, anomaly flags
Vendor agent	Maintenance procedures	Work orders, completion proof
Module advisor	Module manifests, docs, changelogs	Search catalogue, check compatibility, enable on test, request custom module

Safety and control

- Role-based access on every tool call; the real user identity is passed through to the API.
- RAG filters by role, project and unit, so tenants never see owner-only or financial documents.
- Human approval for payments, gate overrides and bulk notices.
- Full audit log of every agent action; consent and data-protection rules for resident data, voice and gate logs.

9. Third-party services we can plug in

Availability and pricing vary by country; confirm each provider before committing. Every provider sits behind an adapter so it can be swapped per country.

Category	Options
Messaging	WhatsApp Business API, Twilio, MSG91 (India), Unifonic (Gulf), Firebase Cloud Messaging, SendGrid or Amazon SES
Payments	Razorpay, Cashfree, PayU, UPI autopay (India); Stripe, Checkout.com, Telr, PayTabs, Network International (UAE); GoCardless (SEPA)
KYC and identity	Sumsub, Onfido, IDfy, Digio, DigiLocker (India), UAE Pass (UAE)
E-signature	DocuSign, Leegality, Zoho Sign
Open banking	Plaid, TrueLayer, Tarabut Gateway
Face, plate, video	AWS Rekognition, Plate Recognizer, Agora or Twilio Video (intercom)
Access hardware SDKs	HID, Hikvision, Dahua, ZKTeco, Suprema
Maps	Google Maps Platform, Mapbox
Accounting and ERP	Zoho Books, Tally, QuickBooks, Xero, SAP, Oracle
CRM	Salesforce, HubSpot, Zoho CRM, LeadSquared
IoT and meters	Tuya, AWS IoT, smart-meter vendor APIs
Documents and OCR	Google Document AI, AWS Textract
AI	Claude API for assistants, ticket triage and document extraction
Monitoring and cloud	AWS, Azure or GCP, Sentry, Firebase Analytics

10. Build roadmap

Phase	Focus	Outcome
1. Foundation	Core Laravel platform, roles and permissions, plugin framework, white labeling and domain mapping, version management, helpdesk, tenants, visitor management	Working base with the highest-volume, lowest-risk features
2. First AI pilot	MCP server over helpdesk and visitor APIs, RAG over bylaws and FAQs, tenant and guard agents	Complaint triage and visitor flow with AI help
3. Money and access	Smart billing, amenities, smart access integration, owner agent	Dues, payments and gate control connected
4. Pre-sale and post-sale	CRM, inventory, agent portal, sales concierge, possession and snag tracking	Revenue-facing features for developers
5. Commercial and scale	Commercial module, analytics, admin agent, searchable module catalogue and module advisor, multi-project rollout	Ready for large developer portfolios

11. Go-to-market plan

- **Pilot first:** run 2-3 live communities (50-500 units) free or low cost to gather usage numbers and testimonials.
- **Right doors at builders:** facility-management and community-management arms, Heads of CRM, Digital/Innovation and Customer Experience; start with mid-tier and regional builders, then approach the large names.
- **Channels:** CREDAI, PropTech summits, Cityscape and Big 5, developer innovation programs, access-control integrators and PMC consultants.
- **Pitch to their problems:** faster collections, cleaner handover, agent and commission control, new convenience-fee revenue, a single branded experience.

- **Pilot proposal:** one tower, 90 days: weeks 1-2 setup and integration, weeks 3-10 live use, weeks 11-12 review against agreed metrics.
- **Enterprise process:** expect vendor registration, security review, ERP and CRM integration and legal terms; plan 6-12 months from first meeting to contract; price per unit per month plus convenience fees.

12. Success measures

- Share of complaints resolved or routed without manual effort.
- Average gate entry time and visitor logging accuracy.
- Dues collected on time, and fewer reminder calls.
- Lead-to-site-visit conversion for the sales concierge.
- Time to launch a new client brand on its own domain.
- Upgrades completed with no regression in other modules.
- Resident satisfaction and adoption by role.

Items in square brackets and pilot targets are placeholders to be agreed with each client.